



DIPLOMA IN COMPUTER TECHNOLOGY

(CLOUD COMPUTING)

DFK51173

CLOUD COMPUTING II

LAB REPORT

Chapter 9: Execute Dockers with Cloud

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JSON

The screenshot shows the AWS Management Console interface for creating a new task definition using JSON. The left sidebar contains navigation links for Amazon Elastic Container Service, including Favourites and recents, Express Mode, Clusters, Namespaces, Task definitions (selected), Daemon task definitions, and Account settings. Below these are links to AWS Resource Explorer, AWS Batch, Amazon ECR, and Repositories. The main content area is titled 'Create new task definition' and includes a description: 'Create or edit a JSON file that defines the container and volume definitions for an Amazon ECS task.' The JSON code is displayed in a dark-themed editor with line numbers 1 through 31. The code defines a task family 'myApp-demo' with two containers: 'simple-app' and 'busybox'. The 'simple-app' container has an image of 'httpd:2.4', 10 CPU, 300 memory, and is essential. It has a port mapping from container port 80 to host port 80 using the 'tcp' protocol. It also has a mount point for a volume named 'my-vol' at the path '/usr/local/apache2/htdocs'. The 'busybox' container has an image of 'busybox', 10 CPU, 200 memory, and is not essential, with an empty entry point. The bottom right of the editor shows '2:16 JSON'.

```
1 {
2   "family": "myApp-demo",
3   "containerDefinitions": [
4     {
5       "name": "simple-app",
6       "image": "httpd:2.4",
7       "cpu": 10,
8       "memory": 300,
9       "essential": true,
10      "portMappings": [
11        {
12          "containerPort": 80,
13          "hostPort": 80,
14          "protocol": "tcp"
15        }
16      ],
17      "mountPoints": [
18        {
19          "sourceVolume": "my-vol",
20          "containerPath": "/usr/local/apache2/htdocs"
21        }
22      ]
23    },
24    {
25      "name": "busybox",
26      "image": "busybox",
27      "cpu": 10,
28      "memory": 200,
29      "essential": false,
30      "entryPoint": [
31
```

The screenshot shows the AWS Management Console interface for the 'myApp-demo:1' task definition. The left sidebar is the same as the previous screenshot. The main content area shows a green success message: 'Task definition successfully created. myApp-demo:1 has been successfully created. You can use this task definition to deploy a service or run a task.' Below the message is the task definition name 'myApp-demo:1' and its last updated time '27 September 2026, 12:30 (UTC+8:00)'. There are buttons for 'Deploy', 'Actions', and 'Create new revision'. Below this is an 'Overview' section with a table of task definition metadata.

Overview			
ARN	Status	Time created	App environment
arn:aws:ecs:us-east-1:437504140493:task-definition/myApp-demo:1	ACTIVE	27 September 2026, 12:30 (UTC+8:00)	EC2
Task role	Task execution role	Operating system/Architecture	Network mode
-	-	-	default
Fault injection			
-			

CREATE TARGET GROUP

Target groups (1/1) Info | What's new?

Filter target groups

Name	ARN	Port	Protocol	Target type
myTargetGroup	arn:aws:elasticloadbalancing...	80	TCP	Instance

Target group: myTargetGroup

Details | Targets | Monitoring | Health checks | Attributes | Tags

arn:aws:elasticloadbalancing:us-east-1:437504140493:targetgroup/myTargetGroup/b2c0fa5d979e1c2a

Target type	Protocol : Port	VPC	IP address type
Instance	TCP: 80	vpc-05361faa9a9152cb2	IPv4

Load balancer
None associated

Total targets	Healthy	Unhealthy	Unused	Initial	Draining
0	0	0	0	0	0

CREATED LOAD BALANCER

Successfully created load balancer: myNLB

It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

myNLB

Details

Load balancer type	Status	VPC	Load balancer IP address type
Network	Provisioning	vpc-05361faa9a9152cb2	IPv4

Scheme	Hosted zone	Availability Zones	Date created
Internet-facing	Z26RNL4JYFTOTI	subnet-0a2b62792d8fa9cef (us-east-1b) and subnet-08a951165181a93c1 (us-east-1a)	September 27, 2026, 12:40 (UTC+08:00)

Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:437504140493:loadbalancer/net/myNLB/835d0fb46bb1c906

DNS name
myNLB-835d0fb46bb1c906.elb.us-east-1.amazonaws.com (A Record)

CREATE CLUSTER

The screenshot shows the AWS Management Console for the Amazon Elastic Container Service (ECS) Clusters page. A blue banner at the top indicates that the cluster 'MyECSClusterX' creation is in progress, with a 'View in CloudFormation' link. The left sidebar shows the navigation menu with 'Clusters' selected. The main content area displays 'Clusters (1)' with a search bar and a table listing the cluster. The table has columns for Cluster, Services, Tasks, and Container Instances. The cluster 'MyECSClusterX' is listed with 0 services, 'No tasks running', and 0 EC2 instances. A 'Create cluster' button is visible in the top right.

Cluster	Services	Tasks	Container Instances
MyECSClusterX	0	No tasks running	0 EC2

The screenshot shows the AWS Management Console for the Amazon Elastic Container Service (ECS) Clusters page, specifically the details for 'MyECSClusterX'. A green banner at the top indicates that the cluster has been created successfully, with a 'View cluster' link. The left sidebar shows the navigation menu with 'Clusters' selected. The main content area displays the cluster details, including the cluster name 'MyECSClusterX', its status 'Active', and various metrics. The 'Cluster overview' section shows the cluster ARN, status, CloudWatch monitoring, action logs, services, tasks, and registered container instances.

Cluster ARN	Status	CloudWatch monitoring	Action logs
arn:aws:ecs:us-east-1:123456789012:cluster/MyECSClusterX	Active	Default	Turned off

Services	Tasks	Registered container instances
0 active 0 draining	0 running 0 pending	1

CONFIGURE CLUSTER LOAD BALANCER

The screenshot shows the 'Create service' page for the 'MyECSClusterX' ECS cluster. The left sidebar contains navigation links for Amazon Elastic Container Service, including Favourites and recents, Express Mode, Clusters, Namespaces, Task definitions, Daemon task definitions, and Account settings. The main content area is divided into sections for VPC, Load balancer type, Container, Load balancer, Listener, and Target group.

VPC
Select the VPC for your load balancing resources.
vpc-05361faa9a9152cb2 (Lab-vpc) [Create a new VPC]

Load balancer type | Info
Specify the load balancer type to distribute incoming traffic across the tasks running in your service.
☐ Application Load Balancer
☒ Network Load Balancer (Selected)

Container
The container and port to load balance the incoming traffic to
simple-app 80:80
Host port: Container port

Load balancer
Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in EC2 Console.
myNLB (myNLB-835d0fb46bb1c906.elb.us-east-1.amazonaws.com) [internet-facing]

Listener | Info
Specify the port and protocol that the load balancer will listen for connection requests on.
☐ Create new listener
☒ Use an existing listener [Listener]

Target group | Info
Specify whether to create a new target group or choose an existing one that the load balancer will use to route requests to the tasks in your service.
☐ Create new target group
☒ Use an existing target group [myTargetGroup]

CREATED CLUSTER

The screenshot shows the 'MyECSClusterX' ECS cluster in the 'Services' tab. A green notification bar at the top states 'myService01X has been deployed successfully.' with a 'View service' button. The cluster details are as follows:

MyECSClusterX (ASG)
Last updated: 27 September 2026, 13:01 (UTC+8:00) [Actions] [Create with Express Mode]

Cluster overview

Cluster ARN arn:aws:ecs:us-east-1:...	Status Active	CloudWatch monitoring Default	Action logs Turned off
Services 1 active 0 draining	Tasks 1 running 0 pending	Registered container instances 1	

Navigation tabs: Services (selected), Tasks, Daemons, Infrastructure, Metrics, Scheduled tasks, Conf

